Technical Report

FastSpeech 2 Text-to-Speech for English (Svarah) and Telugu (IndicTTS)

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Summary of completion status

Item	English	Telugu
Data selection, alignment, feature extraction	Completed	Completed
Model training (10,000-step budget)	Completed (10,000 steps)	Started; 200-step smoke test logged; full run **not completed**
Loss plots, test audio, variance-adaptor plots	Produced	**Not produced** (Because of GPU issue)
Final checkpoint	Step 10000	None

This report analyses the English model in full. For Telugu it documents only the work that was actually executed and states clearly what was not run. No Telugu results are reported or inferred.

1. Implementation approach

The implementation starts from the public PyTorch FastSpeech 2 code base (ming024/FastSpeech2), which follows Ren et al. (ICLR 2021). One copy of the code is kept per language (`English/` and `Telugu/`), each with its own `config.yaml`, so the two models are trained completely separately. Mel-spectrograms are converted to waveforms with the pretrained HiFi-GAN universal vocoder, which is not retrained.

Work was done on Google Colab (GPU). The pipeline for each language is: (1) select and clean data, (2) forced alignment with Montreal Forced Aligner (MFA) to get phone durations, (3) feature extraction (mel, pitch, energy, duration), (4) training, (5) inference with variance-adaptor plots.

Components reused from the reference implementation

- **Encoder / decoder:** feed-forward Transformer (FFT) blocks (`transformer/`).
- **Variance adaptor:** duration predictor with length regulator, pitch predictor and energy predictor, each a small convolutional network (`model/modules.py`). Pitch and energy are predicted at phone level and added to the hidden sequence through learned embeddings.
- **Mel-spectrogram, PostNet, loss and optimizer** (`model/loss.py`, `optimizer.py`): mel loss, PostNet mel loss, pitch, energy and duration losses are summed. The model has about 35.4 M parameters (35,417,409 for English; 35,432,001 for Telugu, which has a different symbol table).
- **Audio and feature code:** STFT, mel filters, pitch with PyWorld (`audio/`, `preprocessor/`).
- **HiFi-GAN** generator for waveform synthesis (`hifigan/`).

Modifications common to both languages

- A single `config.yaml` per language (sections *preprocessing*, *model*, *train*) replaces the original multi-file `config/<dataset>/` layout. `dataset.py` and `preprocess.py` were changed to read it.
- Compatibility fixes for current library versions: `tier._objects` changed to `tier.intervals` (tgt); `torch.load(..., map_location, weights_only=False)`; `model.requires_grad_(False)`; keyword arguments for librosa mel and `pad_center`; pitch, mel and energy cast to float32; corrected import of `to_device`.

- `inference.py` was rewritten to synthesise the test sentences, save one wav per sentence, and save predicted duration, pitch (F0) and energy plots per sentence, with `--restore_step`, `--speaker_id` and `--out` options.
- `plot_losses.py` creates training and validation loss plots from the logs.

English-specific changes

- Text front end: lowercase letters only, numbers expanded with *inflect*, out-of-vocabulary words converted to ARPAbet with *g2p_en* (CMU dictionary).
- MFA dictionary extended with pronunciations from the MFA G2P model for 1,160 out-of-vocabulary word types found in the Svarah transcripts.
- Optional `--fix_heteronyms` flag for test sentence E2 (conduct as noun and as verb); it is off in the default results.

Telugu-specific changes (implemented; training not completed)

- Telugu text cleaning (NFC, only Unicode block U+0C00 to U+0C7F, no digits or Latin letters in training text).
- **Grapheme-level symbols:** no Telugu pronunciation dictionary was available, so each Telugu character is treated as one phone (for example *kaalam* is written as its characters separately). 63 symbols found in the alignments are stored in `text/telugu_phones.json` and added to `symbols.py` with an `@` prefix.
- An MFA acoustic model was **trained from scratch** on the selected Telugu data (monophone and triphone stages, 20 iterations each) because no pretrained Telugu aligner was used.
- `preprocess.py` replaced by a version that reads `config.yaml` and goes directly from the prepared wav/lab/TextGrid files to feature extraction.
- Numbers, dates and times (sentence T4) are meant to be converted to Telugu words at inference time. **This code path was not executed.**

2. Datasets, preprocessing and training configuration

Both languages use the same recipe: pick about 5 h for training and about 1 h for validation with a fixed random seed (42), resample to 22.05 kHz mono, peak-normalise to 0.95, and write one wav and one lab file per utterance. Training and validation never share an utterance, and a transcript used in training is never used in validation (both checks were run and returned empty sets).

	English: Svarah	Telugu: IndicTTS Telugu
Source	AI4Bharat, Hugging Face *ai4bharat/Svarah* (Indian-accented English)	IITM Speech Lab, IIT Madras, Hugging Face *SPRINGLab/IndicTTS_Telugu*
Raw size	About 9.6 h (all splits pooled)	8,576 utterances, about 36.7 h, 48 kHz; 1 female and 1 male speaker
Filters	2 to 15 s; drop transcripts with 6+ digit strings and million/billion/trillion/quadrillion	2 to 15 s; drop any sentence with digits or Latin letters
Usable after filtering	4,452 utts, 7.32 h	5,205 utts, 11.79 h
Text cleaning	Remove tags, expand numbers, lowercase, keep a-z, apostrophe, space	NFC, remove zero-width joiners, keep Telugu characters and spaces only
Speakers	8 gender x age-group labels (group labels, not single speakers)	2 (Female, Male); equal time budget per speaker
Selected train / val	5.00 h (3,056 clips) / 1.00 h (613 clips)	5.00 h (2,110 clips) / 1.00 h (423 clips)
Aligner	MFA pretrained *english_us_arpa* + G2P for OOV words	MFA trained from scratch, grapheme dictionary (12,545 word types)
Clips after alignment and preprocessing	3,660 of 3,669 selected (3,047 train, 613 val)	2,510 of 2,533 TextGrids (23 missing: 20 train, 3 val)
After preprocessing	3,047 train (about 3.89 h), 613 val (about 0.79 h)	2,090 train (about 3.44 h), 420 val (about 0.69 h)
Pitch mean / std (Hz)	178.9 / 57.5	197.9 / 45.5
Training status	10,000 steps completed	Not completed

Training configuration (English)

Setting	Value
Steps	10,000 (the reference default was 30,000; reduced to fit available Colab GPU time)
Batch size	16 (190 batches per epoch on 3,047 utterances)
Parameters	35,417,409
Speed on Colab GPU	about 2.2 iterations/s; the last 4,000 steps took about 35 min
Checkpoints	Saved to Google Drive every 2,000 steps; training was resumed from step 200 and step 6000 after disconnects
Validation	Run every 1,000 steps on the 613-utterance validation set
Vocoder	HiFi-GAN universal (not fine-tuned)
Inference speaker	Speaker id 2 (Female-30-45)

Telugu: the 200-step smoke test was run with the same settings as English apart from the symbol table (35,432,001 parameters). Its logged losses were train total 8.53 and validation total 7.88 at step 200. The saved notebook contains no logs from the later full-training attempt, so no further training numbers are reported.

3. Training and validation loss analysis

3.1 English



Figure 1. English training loss (total and components), logged every 100 steps.

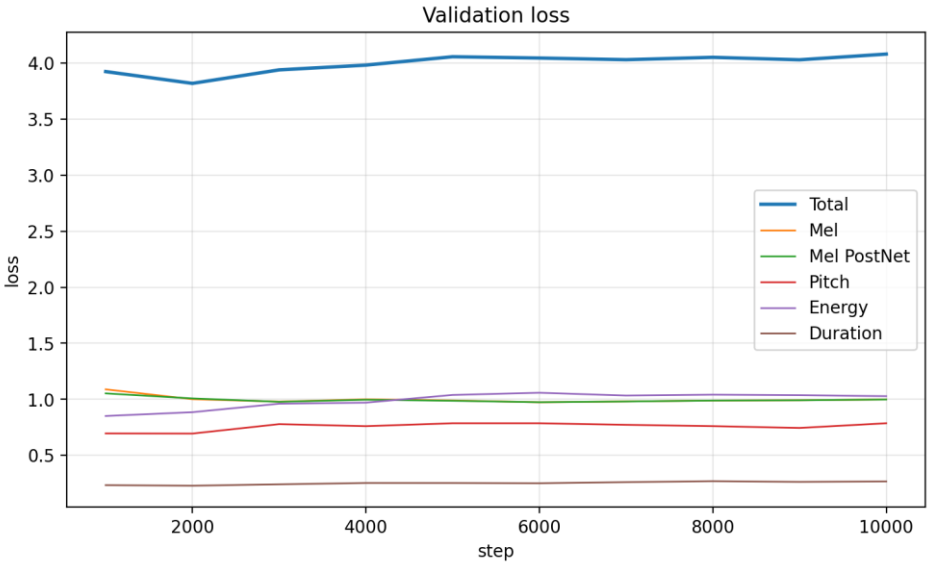


Figure 2. English validation loss, evaluated every 1,000 steps.

Step	Train total	Val total	Val mel	Val pitch	Val energy	Val duration
1,000	5.07	3.92	1.09	0.70	0.85	0.23
2,000	3.31	**3.82 (lowest)**	1.00	0.69	0.89	0.23
4,000	2.57	3.98	1.00	0.76	0.97	0.25
6,000	1.82	4.04	0.97	0.79	1.06	0.25
8,000	1.69	4.05	0.99	0.76	1.04	0.27
10,000	1.49	4.08	1.00	0.79	1.03	0.27

Values are the logged training/validation numbers from the training log (training values are single-batch logs and therefore noisy).

Convergence behaviour

- **Training loss** decreases steadily and smoothly from about 4.4 (first plotted point) to about 1.49 at step 10,000, with no divergence or loss spikes. It had not flattened at step 10,000, so more steps

would continue to lower it. Mel and PostNet mel losses fall slowly (about 1.3 to 0.61); pitch, energy and duration losses fall faster and reach about 0.10, 0.10 and 0.06.

- An early transient is visible around step 1,000: the logged pitch loss was high (1.74) while the pitch predictor was still settling, after which it dropped to about 0.25 by step 3,000.
- **Validation loss** is lowest at step 2,000 (3.82), then rises slightly and plateaus around 4.03 to 4.08 from step 5,000 onward (about 7% above the minimum).

Overfitting

There is a clear generalisation gap. At step 10,000 the training total loss (1.49) is much lower than the validation total loss (4.08, about 2.7 times higher), and the gap grows over time because the training loss keeps falling while the validation loss stays flat or rises. The model starts to overfit after roughly 2,000 steps. The gap is largest for the variance predictors: validation pitch loss is 0.79 against 0.10 in training, validation energy loss 1.03 against 0.10, and validation duration loss 0.27 against 0.06. Validation mel loss (about 1.0 against 0.61) is much closer.

Likely causes: (i) only about 3.9 h of aligned training audio spread over 8 speaker groups (roughly 30 min per group), so the pitch and energy predictors memorise per-utterance patterns; (ii) Svarah is spontaneous, accented speech with variable recording conditions, so pitch and energy are inherently harder to predict from text than for read speech; (iii) MFA alignments on this type of speech are less accurate, which adds noise to the duration targets. The validation total did not diverge, so the model remains usable, but the plateau shows that longer training on the same data yields little generalisation benefit. Regularisation, more data or speaker-grouped training would be the next steps.

Note on checkpoints: the lowest validation loss is at step 2,000, but sample quality was compared at steps 2,000, 6,000 and 10,000 and step 10,000 was kept as the final model because it had the full training budget and the lowest mel and training losses.

3.2 Telugu

No Telugu loss curves are available. Only the 200-step smoke test was logged (train 8.53, validation 7.88 at step 200), which shows that preprocessing, dataset loading and training start correctly but says nothing about convergence or overfitting. The plotting step (notebook section 30) was not run.

4. Observations on the generated speech

Test sentences were synthesised with the step-10,000 English model, speaker id 2, HiFi-GAN universal vocoder. The table gives what the text front end does for each sentence and the expected difficulty. Cells highlighted in yellow are for the author to complete after listening to the files in `Results/English/test_generated_audios/`; they are not filled in here because audio was not assessed in preparing this report.

ID	Challenge in the sentence	Front-end handling	Listening notes (pronunciation, quality)
E1	Very long rare word (*antidisestablishmentarianism*) plus long clause	g2p_en produces a 29-phone ARPAbet sequence for the long word (visible in the duration plot): AE N T AY D IH S AH S T AE B L IH SH M AH N T EH R IY AH N IH Z AH M. Total sentence is about 80 phones.	The long word is hard to follow; the syllables [*-lish-ment-*] are merged or slurred
E2a	Heteronym *conduct* as a noun (stress on first syllable)	Default g2p resolves by part of speech; result depends on the tagger. `--fix_heteronyms` run was also made to compare.	*conduct* is K AA1 N D AH0 K T, stressed on the first syllable, which is correct for a noun. The pitch peaks at 270 Hz on its first syllable. The phrase *during the* looks weak: one 348 ms vowel, flat 173 Hz pitch and energy near 5 to 10.
E2b	Heteronym *conduct* as a verb (stress on second syllable)	As above.	conduct *has the same phones as E2a, so the verb stress was* ***not*** *produced. The energy pattern is inverted: the stressed vowel is quiet and the unstressed one is loud.*
E3	Typo preserved as given: "rhyth m" (word split by a space)	The text becomes two tokens, *rhyth* and *m*; neither is a normal word, so g2p guesses pronunciations (a spelled letter name for *m* is possible).	*rhyth* is read as “ryth” and *m* as the letter name “em”, so the phrase is not “rhythm”. The extra “m” gets its own pitch peak and high energy
E4	Time, date and year ("10:45 AM", "12 November 2026") plus a nonsense word *andaroo*	Digits are expanded to words (inflect); *AM* and *andaroo* are handled by g2p. Number reading style (for example "twenty twenty-six" vs "two thousand twenty-six") depends on the expansion rules.	The time, date and year are expanded correctly: “ten forty five”, “twelve”, “twenty twenty six”. *AM* is spelled as letters. The year is rushed (35 ms phones, near-zero energy on the first “twenty”). The sentence ends with an unusual pitch rise (182 to 208 Hz)

General observations that follow from the training results

- Because mel validation loss plateaus near 1.0, the output spectrogram is smoothed compared with natural speech, which is typical of this loss at this data size; any buzzy or muffled quality comes from this and from the universal vocoder not being fine-tuned on Svarah audio.
- The speaker is a demographic group label (Female-30-45) rather than a single voice, so the synthesised voice is an average of several speakers in that group.

Telugu: no audio was generated (inference notebook sections 31 to 33 were not run), so there are no observations for T1, T2a, T2b, T3 or T4.

5. Variance-adaptor analysis

The variance adaptor predicts, for each input phone, a duration (in frames of 256 samples = 11.6 ms), a pitch value (F0 in Hz) and an energy value. The plots below are the model's predictions for sentence E1 at step 10,000. They are piecewise constant because each phone gets one predicted value.

5.1 E1: duration

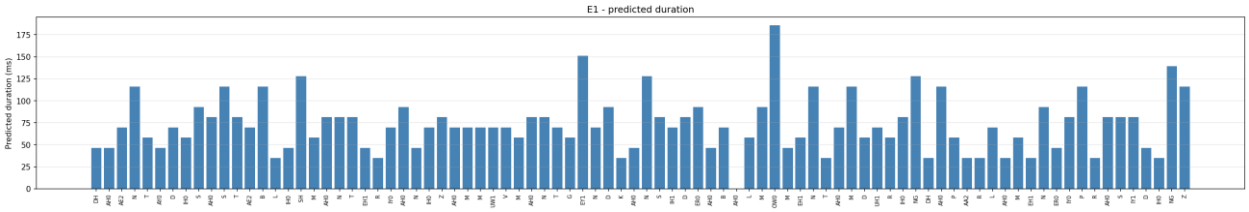


Figure 3. E1 predicted duration per phone.

- Durations lie between about 35 ms and 185 ms and are multiples of the 11.6 ms frame (35, 46, 58, 69, 81, 93, 116, ...), as expected from integer frame durations.
- The longest phones are stressed or word-final vowels and nasals: the first vowel of *momentum* (about 185 ms), the vowel in *gained* (about 150 ms), and the final *-ing* and *-s* of *proceedings* (about 140 and 116 ms). The pattern reflects phrase-final lengthening and stress.
- The shortest phones (about 35 to 46 ms) are reduced vowels (AH, IH) in unstressed syllables, for example inside *antidisestablishmentarianism* and *considerable*, and short function words (*the*), matching natural reduction.
- Inside the very long word the durations are fairly even (about 45 to 115 ms per phone), so the model spreads time across the 29 phones rather than rushing them, which supports intelligibility of this word.

5.2 E1: pitch (F0)

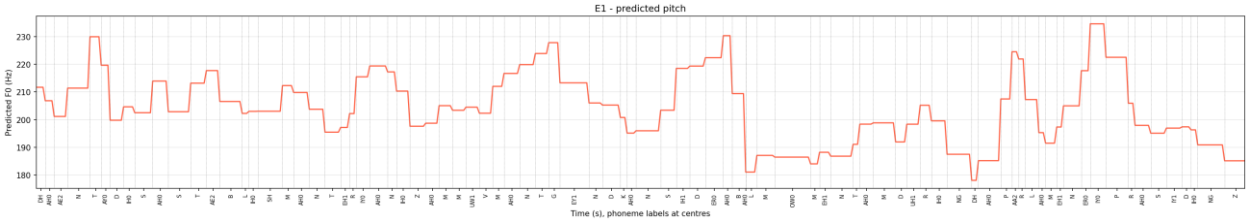


Figure 4. E1 predicted pitch.

- Predicted F0 ranges from about 178 Hz to 235 Hz. This matches a female speaker group (the dataset mean is about 179 Hz) and a narrower range than the training spread (std about 57 Hz), showing that the predictor stays near the speaker mean, a known effect of regression with a mean-squared-error loss.
- There is a gradual downward trend over the sentence (declination): about 200 to 230 Hz in the first words, about 185 to 190 Hz around *momentum* and *during*, and about 185 to 190 Hz at the end, a falling contour typical of a statement.
- Local peaks fall on stressed syllables: *anti-* (T, about 230 Hz), *gained* (about 228 Hz), *considerable* (ER, about 230 Hz) and the highest peak (about 235 Hz) on the stressed syllable of *parliamentary*, followed by a fall to the end. Pitch drops sharply at the boundary between the two clauses before *the parliamentary proceedings*, which marks the phrase break.

5.3 E1: energy

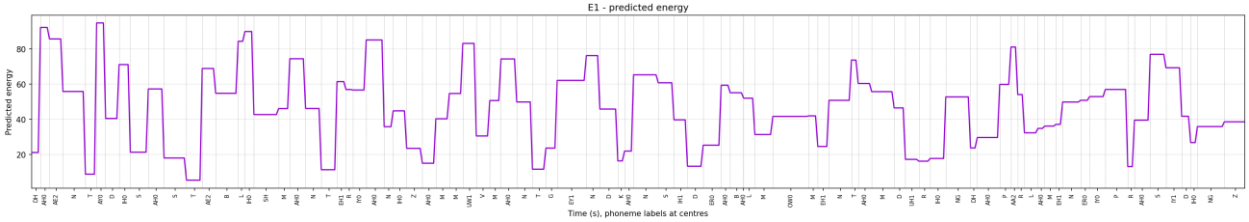


Figure 5. E1 predicted energy.

- Energy varies from about 5 to 95. Vowels and sonorants (AH, AY, IH, UW, AE) are predicted loud (about 70 to 95), while stops and fricatives (T, D, S, K, closures) are quiet (about 5 to 25), which follows the natural pattern of the data.
- The first vowels of the sentence are loudest (about 90 to 95) and the energy decays toward the end (final phones about 35 to 40), matching typical energy decay in a declarative sentence.
- Energy depends mostly on phone class and only weakly on lexical stress (for example unstressed AH can be as loud as stressed vowels). This is in line with the validation energy loss being high (about 1.0), meaning the model learned phone-class energy but not fine prosodic emphasis.

5.4 E2a, E2b, E3, E4

Variance-adaptor plots for these sentences are saved in

`Results/English/variance_adaptor_plots/{duration,pitch,energy}/`.

ID	What to look for in the plots
E2a / E2b	Compare the duration and pitch of *conduct* in the noun versus the verb context: the noun should have a stress peak on the first syllable, the verb on the second. If the two plots look identical for *conduct*, the heteronym was not distinguished.
E3	Look at the phones assigned to *rhyth* and *m*: an extra short phone or a long pause between them confirms the space typo changed the phone sequence.
E4	Look at the long run of number words (time and year): expect many short phones with flat pitch (list-like reading), and check pause lengths around *AM* and *andaroo*.

Telugu: no variance-adaptor plots were generated (the plots require a trained checkpoint and the inference step, which were not run).

6. Comparative observations, limitations and improvements

Because no Telugu model was evaluated, the comparison below is between the two pipelines as built, not between synthesised outputs.

Aspect	English	Telugu
Front end	g2p_en + CMU dictionary; ready-made pronunciation resources and a pretrained MFA model	No lexicon or g2p; characters used as phones; aligner trained from scratch
Alignment quality	Reliable on read-style English but spontaneous Svarah speech loses some clips (3.89 of the 5 h selected survived)	Studio-quality read speech aligned well (2,510 of 2,533), but 3.44 of 5 h survived feature extraction and alignment quality was not inspected
Speakers	8 demographic groups; more variability, harder pitch/energy prediction	2 studio speakers; cleaner data, easier to learn
Numbers / digits	Expanded by inflect at training and inference	Excluded from training; must be expanded to Telugu words at inference (not tested)
Outcome	Trained 10,000 steps, overfits after about 2,000 steps	Not trained to completion; no results

Limitations and failures

- **Telugu was not completed.** The main reasons were limited GPU availability on Colab and limited time, and because of the work, time during the assignment. Telugu needed extra setup that English did not (training an aligner from scratch, fixing library compatibility problems, restoring data after Colab resets), and the GPU quota ran out before full training finished. As a result the checkpoint, loss plots, test audio and variance-adaptor plots for Telugu are missing and the Telugu test sentences T1 to T4 are untested.
- English overfits (Section 3): validation loss is lowest at step 2,000 and the pitch/energy predictors generalise poorly.
- Only about 3.9 h of aligned English audio from 8 speaker groups; each group has under about 30 minutes.
- The English test sentences are deliberately hard: a very long rare word (E1), a heteronym (E2), a typo with a space (E3), and a time/date with a nonsense word (E4). These depend on the g2p front end rather than the acoustic model.
- Vocoder is not fine-tuned on either dataset.

Possible improvements

- Finish Telugu training and run the full evaluation, then compare against English using the same test-sentence analysis.
- English: use early stopping or pick the checkpoint by validation loss, add dropout/weight decay, add more data or merge speaker groups, and use a pronunciation dictionary fix list for heteronyms (E2) and number/time expansion rules (E4).
- Telugu: a rule-based grapheme-to-phoneme step (schwa deletion, gemination, loanword handling) instead of one phone per character, and a Telugu number-to-words normaliser for dates and times (T4).
- Fine-tune the HiFi-GAN vocoder on the training audio of each language, and evaluate with objective metrics (for example MCD, F0 RMSE) and a listening test.

7. Reproducibility

Software

Python 3 on Google Colab with a GPU. Python packages (see `requirements.txt`): torch, numpy, scipy, pandas, scikit-learn, librosa, soundfile, pyworld, tgt, pyyaml, matplotlib, tensorboard, tqdm, inflect, unidecode, g2p_en, nltk, pypinyin, datasets<4, huggingface_hub. Montreal Forced Aligner is installed with conda: `conda install -c conda-forge montreal-forced-aligner`. The HiFi-GAN universal generator (`generator_universal.pth.tar`) must be placed in `hifigan/`. NLTK data for g2p_en: averaged_perceptron_tagger_eng, averaged_perceptron_tagger, cmudict.

Configuration files: `English/config.yaml` and `Telugu/config.yaml` (data paths, model sizes, optimizer, 10,000 total steps, checkpoint and log paths).

English commands (all executed)

```
cd English
# 1. wav + lab from ai4bharat/Svarah (needs a Hugging Face token); seed 42, 5 h train / 1 h val
# 2. alignment
mfa model download acoustic english_us_arpa
mfa model download dictionary english_us_arpa
mfa model download g2p english_us_arpa
mfa g2p oov_words.txt english_us_arpa oov_pron.dict
cat oov_pron.dict >> svarah_extended.dict
mfa align --clean -j 2 raw_data/Svarah svarah_extended.dict english_us_arpa \
    preprocessed_data/Svarah/TextGrid
# 3. features
python preprocess.py config.yaml
# 4. training (resume with --restore_step N)
python train.py config.yaml
# 5. loss plots
python plot_losses.py config.yaml ../Results/English
# 6. inference + variance-adaptor plots
python inference.py config.yaml --restore_step 10000 --speaker_id 2 --out ../Results/English
python inference.py config.yaml --restore_step 10000 --speaker_id 2 --fix_heteronyms \
    --out ../Results/English_extra/heteronym_fix
```

Telugu commands (steps 1 to 4 were executed; step 4 not finished; steps 5 and 6 not run)

```
cd Telugu
# 1. wav + lab from SPRINGLab/IndicTTS_Telugu (IITM IndicTTS); seed 42, 2.5 h + 2.5 h train,
#    0.5 h + 0.5 h val per speaker
# 2. grapheme dictionary (word <TAB> c h a r s) and aligner trained from scratch
mfa train --clean -j 2 --config_path mfa_config.yaml \
    --output_directory telugu_textgrid raw_data/Telugu telugu_dict.txt telugu_acoustic.zip
# 3. features
python preprocess.py config.yaml
# 4. training (launched, not completed)
python train.py config.yaml          # resume: --restore_step N
# 5-6. NOT RUN
python plot_losses.py config.yaml ../Results/Telugu
python inference.py config.yaml --restore_step 10000 --speaker_id 0 --out ../Results/Telugu
```

Data and checkpoints: the English checkpoint is `English/checkpoints/10000.pth.tar` (see the README for the download link). No Telugu checkpoint is provided. The Telugu dataset is the IndicTTS database from the IITM Speech Lab, IIT Madras .

References

Y. Ren et al., FastSpeech 2: Fast and High-Quality End-to-End Text to Speech, ICLR 2021. Reference code: github.com/ming024/FastSpeech2. Kong et al., HiFi-GAN, NeurIPS 2020. McAuliffe et al., Montreal Forced Aligner, Interspeech 2017.